

Reading and Plotting Points on a Grid

Answer Key

Grades 4–5 Mathematics

Quick Answers

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|---|--|
| 1. x-coordinate of L = 2, y-coordinate of L = 3 | 8. (a) x-coordinate after 4 steps = 4, (a) y-coordinate after 4 steps = 9, — |
| 2. P | 9. (a) mean of the y-coordinates = 14 cm, (b) range of the y-coordinates = 12 cm |
| 3. x-coordinate landed on = 6, y-coordinate landed on = 8 | 10. (a) y-coordinate from rule 1 = 8, (b) y-coordinate from rule 2 = 10, (c) the x where both rules agree = 6, — |
| 4. 4 units | |
| 5. — | |
| 6. y-coordinate: height at hour 5 = 5 | |
| 7. (a) length of side DE = 7 units, — | |
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What is verified

6 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 7, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

5.G.A.1–A.2 – *problems 1, 2, 3, 4, 5, 7*

5.MD – *problem 9*

5.OA.B.3 – *problems 6, 8, 10*

Difficulty 1–5 of 5 across 19 tagged checks.

- 1.** Use the Park Map. Write the ordered pair for landmark *L*.

Solution: Start at the origin and count across to *L* first: 2 squares. Then count up: 3 squares. The across number is written first.

(2, 3)

- 2.** Use the Park Map. Which labelled landmark is farthest to the right?

Solution: “Farthest to the right” means the largest *x*-coordinate, so compare only the across numbers: *L* is 2, *Q* is 3, *M* and *N* are 6, and one landmark is 9.

P

- 3.** Use the Park Map. Start at landmark *L*, move 4 units right, then 5 units up. Write the ordered pair where you land.

Solution: L is at $(2, 3)$. Moving right changes only the across number: $2 + 4 = 6$. Moving up changes only the up number: $3 + 5 = 8$. That lands on landmark N . $(6, 8)$

4. Use the Park Map. Landmarks L and M sit on the same horizontal line. How far apart are they?

Solution: Both landmarks have the same up number, so the gap is the difference of their across numbers: $6 - 2$. Count the squares between them on the map to check. 4 units

5. Plot and label $A(1, 4)$, $B(5, 4)$ and $C(5, 9)$ on the grid.

Solution — instructor-judged (a drawing). A : one square right of the origin, then four up. B : five right, four up — level with A . C : five right, nine up — directly above B .

Check each mark by counting across first and then up. The mistake to look for is a point placed at $(4, 1)$ instead of $(1, 4)$: that is the reversed-order error, and it is the single most common one on this skill. All three letters should be written beside their dots.

6. A candle is 20 centimetres tall and burns down 3 centimetres each hour. Write the ordered pair for hour 5.

Solution: The across number is the hour, which is 5. The up number is the height after 5 hours of burning: $20 - 3 \times 5 = 20 - 15$. $(5, 5)$

The two 5s here mean completely different things — one counts hours across, one counts centimetres up.

7. Plot $D(2, 2)$, $E(9, 2)$, $F(9, 6)$, $G(2, 6)$ and join them in order. (a) How long is side DE ?

Solution (a): D and E have the same up number, 2, so DE is horizontal and its length is the difference of the across numbers: $9 - 2$. 7 units

(b) — instructor-judged (a drawing). The four points form a rectangle 7 units wide and 4 units tall, with DE along the bottom and GF directly above it. Check that D and E sit on the same horizontal line and that G is directly above D ; freehand segments are fine as long as the figure closes back to D .

8. A rule says: start at $(0, 1)$, and for every step move 1 unit right and 2 units up. (a) Ordered pair after 4 steps. (b) Plot the first three points.

Solution (a): Each step adds 1 to the across number, so after 4 steps it is $0 + 1 \times 4 = 4$. Each step adds 2 to the up number, so it is $1 + 2 \times 4 = 9$. $(4, 9)$

(b) — instructor-judged (a drawing). The first three points are $(0, 1)$, $(1, 3)$ and $(2, 5)$. Check that every mark is one square right and two squares up from the one before it, and that the three marks fall on a straight line — a constant rule always plots straight.

9. The candle points for hours 0 to 4 are $(0, 20)$, $(1, 17)$, $(2, 14)$, $(3, 11)$, $(4, 8)$. (a) Mean of the five y -coordinates. (b) Range of the five y -coordinates.

Solution (a): Add the up numbers: $20 + 17 + 14 + 11 + 8 = 70$. Divide by how many there are: $70 \div 5$. 14 cm

(b) The range is the largest minus the smallest up number: $20 - 8$. 12 cm

Notice the mean, 14, is also one of the listed heights. That happens because the heights drop by the same amount each hour, so the middle value and the mean coincide.

10. Rule 1: the y -coordinate is double the x -coordinate. Rule 2: the y -coordinate is 6 more than the x -coordinate. (a) Rule 1 at $x = 4$. (b) Rule 2 at $x = 4$. (c) Where do the rules agree? (d) Explain how you found it.

Solution (a): Double 4. 8

(b) Six more than 4. 10

(c) Extend both lists of ordered pairs:

Rule 1: $(4, 8)$, $(5, 10)$, $(6, 12)$. Rule 2: $(4, 10)$, $(5, 11)$, $(6, 12)$.

They first agree at the point $(6, 12)$, so the x -value is 6

(d) — instructor-judged. Look for a stated method, not just the number. Either is fine: extending both lists of ordered pairs until the up numbers match, or noticing that rule 1 starts 6 behind at $x = 4$ but gains one extra unit on every step, so it catches up after 2 more steps.